SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category – I by UGC

Seat No. | | | | | | |

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2022-26
Semester: I
Course: Programming and Problem Solving
Course Code: 0705210114

Date: 24/12/2023
Day: Sunday

Maximum Marks: 30
Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Make suitable assumptions wherever necessary and figures to the right indicate full marks.

Q.1	a)	Write a Python program to prompt the user to enter a number, multiply and divide the number by 4 without using the arithmetic operators and display the output.	3	CO1, BL3
	b)	Describe in detail computational thinking with example.	4	CO1, BL2
Q.2	a)	Write a short note on break and continue statement with example.	3	CO2, BL3
	b)	Write a python program to prompt a number from the user and find whether the given number is multiple of 5 and 8 or not using ternary operator.	4	CO2, BL3
Q.3	a)	Write a python program to create a function containing list of five friends. The function should search for the friend's name, if the name is not in the list, add the friend's name in the list and returns the new list or else returns the list as it is.	4	CO3, BL3
	b)	Write the advantages of using lambda function over the user-defined functions.	4	CO3, BL3
Q.4	a)	Write a program to create a class Flower with attributes name, color, and behavior show.	4	CO4, BL3
	b)	Differentiate between constructors and destructors of python in a class.	4	CO4, BL4

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SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

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Re-accredited by NAAC with 'A' Grade (3.58/4) Awarded Category – I by UGC

Seat No. | | | | | | |

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2022-26
Semester: I
Course: Programming and Problem Solving
Course Code: 0705210114

Batch: 2022-26 (Sem I)
Regular October/Odd 2022 Examination

Date: 23/01/2023
Day: Monday

Maximum Marks: 30
Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

Q.1	a)	Design an algorithm for finding the sum of digits of a 3-digit number.	3	CO1, BL3
	b)	Write a short note on Data Representation in computers.	3	CO1, BL2
Q.2	a)	Create a Dictionary with 5 elements in python with keys as "courses" and values as "number of students enrolled". Display all values and keys separately.	3	CO2, BL3
	b)	Write a python program to check if a 3-digit number entered by the user is Armstrong number or not.	3	CO2, BL3
Q.3	a)	Write a program to find the factorial of a number using a recursive function.	4	CO3, BL3
	b)	Describe keyword argument and default argument with an example.	4	CO3, BL2
Q.4	a)	Explain the usage of constructor and destructor in python class.	4	CO4, BL1
	b)	Write a program to create a base class BIRD with attributes such as color, weight, and behavior Fly. Create another two child classes peacock and eagle with the attribute of flying height and method to print flying height. Explain Method Overriding.	6	CO4, BL3

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Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2022-26
Semester: I
Course: Programming and Problem Solving
Course Code: 0705210114

Batch: 2022-26 (Sem I)
Backlog April/Even 2023 Examination

Date: 26/06/2023
Day: Monday

Maximum Marks: 30
Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

Q.1	a)	Design an algorithm to find whether a number is odd or even.	3	CO1, BL3
	b)	Write a short note on Computational Thinking.	3	CO1, BL2
Q.2	a)	Create a list of 5 friends in python. Display the names of all friends along with their length of the names separately.	3	CO2, BL3
	b)	Write a program to create a game such that, when a player enters his date of birth, the program computes single digit sum of it and displays a message based on the sum as follows: if sum is 1, 2 or 3: You are Intelligent, 4, 5 or 6: You are Cool, 7, 8 or 9: You are Gorgeous.	3	CO2, BL3
Q.3	a)	Write a program to find the multiplication table of a number using a recursive function.	4	CO3, BL3
	b)	Write a short note on modules in python and show the use of any predefined module.	4	CO3, BL2
Q.4	a)	With help of an example, demonstrate the use of "__init__" (constructor) and "__del__" (destructor) in python class.	4	CO4, BL4
	b)	Write a program to create a base class STUDENT with attributes such as name, age, and behavior display. Create a child class BTechStudent with the attribute of semester and method to print the student's name, age and semester. Explain Inheritance.	6	CO4, BL3

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Batch: 2022-26 (Sem I)
Backlog April/Even 2023 Examination

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Seat No. | | | | | | |

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY(COMPUTER SCIENCE AND ENGINEERING)

Batch: 2021-25,2022-26,2023-27
Semester: II
Course: Programming and Problem Solving
Course Code: 0705210206

Date: 30/11/2024 Maximum Marks: 30

Day: Saturday Time: 13:30 - 14:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

a) **Use Ball Pen/ Ink Pen (Blue only) to answer all questions.**
b) **Write your seat number in words and figures in the Answer sheet provided to you.**
c) **Figures to the rights indicate maximum marks. All questions are compulsory.**

Q1.	a)	Construct a flowchart algorithm that takes three coefficients a,b and c of a quadratic equation ($ax^2+bx+c=0$) as input and computes all possible roots.	3	CO1, BL3
	b)	Demonstrate the concept of computational thinking using a real-world scenario.	3	CO1, BL2
Q2.	a)	Explain string indexing and slicing in Python with examples.	4	CO2, BL2
	b)	Write a Python program to accept three digits and print all possible combinations from the digits.	4	CO2, BL3
Q3.	a)	Write a program to find whether a number entered by the user is an armstrong number or not using user-defined function. (Hint: armstrong no is a no that is equal to the sum of its own digits raised to the power of the number of digits. e.g 153 is an armstrong no $1^3+5^3+3^3=1+125+27=153$)	4	CO3, BL3
	b)	Explain the process of creating and importing a user-defined module in Python.	4	CO3, BL2
Q4.	a)	What are attributes and methods in Python classes? How are they defined and accessed?	3	CO4, BL1
	b)	Write a Python program to model different geometric shapes using inheritance. Define a base class Shape with attributes like length, width, and radius. Then, create subclasses Rectangle and Circle that inherit from the Shape class and display the area of each shape.	5	CO4, BL3

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Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Seat No.

Programme: (070521) BACHELOR OF TECHNOLOGY(COMPUTER SCIENCE AND ENGINEERING)

Batch: 2021-25,2022-26,2023-27
Semester: II
Course: Programming and Problem Solving
Course Code: 0705210206

Date: 28/05/2024

Day: Tuesday

Maximum Marks: 30

Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

a) **Use a Ball Pen/ Ink Pen (Blue only) to answer all questions.**
b) **Write your seat number in words and figures in the Answer sheet provided to you.**
c) **Figures to the right indicate maximum marks. All questions are compulsory.**

Q1.	**a)**	Construct a flowchart to determine whether a given number is prime or not. Prompt the user to input an integer, apply the prime number test, and display the result.	3	CO1, BL3
	b)	Demonstrate the concept of computational thinking using a real-world scenario.	3	CO1, BL2
Q2.	**a)**	What is a dictionary in Python? Explain any three dictionary methods.	4	CO2, BL2
	b)	Write a Python program to remove punctuations from a string entered by the user without using string inbuilt methods.	4	CO2, BL3
Q3.	**a)**	Write a program to find whether a number entered by the user is a spy number or not using user-defined function. (Hint: spy number is a number if the sum and product of its digits are equal e.g 132)	4	CO3, BL3
	b)	What is a module in Python? How can you import specific attributes or functions from a module in Python?	4	CO3, BL1
Q4.	**a)**	What is a class in Python? How does it serve as a blueprint for creating objects?	3	CO4, BL1
	b)	Write a Python program to model different geometric shapes using inheritance. Define a base class Shape with attributes like length, width, and radius. Then, create subclasses Rectangle and Circle that inherit from the Shape class and display the area of each shape.	5	CO4, BL3

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Seat No. | | | | | | |

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2021-25,2022-26
Semester: II
Course: Programming and Problem Solving
Course Code: 0705210206

Date: 26/12/2023 Maximum Marks: 30

Day: Tuesday Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. **Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

Q.1	a)	**Design a flowchart to find greatest of three numbers.**	**3**	**CO1, BL3**
	b)	**Write a short note on IDE and their common features.**	**3**	**CO1, BL2**
Q.2	a)	**What is typecasting? Explain its different types with examples.**	**4**	**CO2, BL2**
	b)	**On a birthday party, there are some family members and friends to be invited. Family members are sure to be invited and their name cannot be changed whereas, there are some friends whose name could be updated. Write a python program containing a single record of all the invitees having 5 family members and list of 5 friends. Later modify one of the friends name and display the members.**	**4**	**CO2, BL3**

Q.3	a)	Write a python program that accepts vehicle registration number from the user and prints whether the given input is valid or not. The vehicle registration number must be in 'AABBCCDDDD' format where, AA - state code(alphabets) BB- RTO city code(numbers) CC- series (alphabets) DDDD- numbers. (Hint : Use string inbuilt methods).	4	CO3, BL3
	b)	Write a program to differentiate between map and filter functions in python.	4	CO3, BL3
Q.4	a)	Illustrate the use of constructors and destructors in class.	4	CO4, BL4
	b)	Write a program to create a base class Flower with attributes such as name, color, and behavior show. Create a child class Rose with the attribute of petalcount and method to print the Flower's name, color and petalcount.	4	CO4, BL3

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Seat No. | | | | | | |

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2021-25
Semester: II
Course: Programming and Problem Solving
Course Code: 0705210206

Batch: 2021-25 (Sem II)
Backlog October/Odd 2022 Examination

Date: 09/01/2023

Day: Monday

Maximum Marks: 30

Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. **Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

Q.1	**a)**	Illustrate different phases of Computational thinking with Example.	3	**CO1, BL4**
	b)	Write an algorithm to find factorial of a number using recursion.	3	**CO1, BL3**
Q.2	**a)**	Predict the output and justify your answer: **i)** -67%7 **ii)** 6.6//6 **iii)** (90-70) * 10/5 **iv)** not False **v)** type((4,5)) **vi)** 5*1**2	3	**CO2, BL3**
	b)	Describe any two list operations and list methods. Write a Python program to check whether an element is present in the list or not using membership operator.	3	**CO2, BL2**
Q.3	**a)**	Write a Python program to prompt a number from user and find reverse of number using recursion	4	**CO3, BL3**
	b)	Discuss about any two methods for passing parameters to a function with examples.	4	**CO3, BL2**

Q.4	**a)**	Create a class named School having class attribute as SchoolCity. Instance attributes as StudName and StudStd. The Student class should have a method as SetAge(), that prompts age from user and displays the age.	5	**CO4, BL5**
	b)	Create a child class that will inherit all the features of the School class used in Q.4 (a) and have additional attribute as SchoolBranch. Also, create show() method to display all the features of School class.	5	**CO4, BL5**

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Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2022-26, 2021-25
Semester: II
Course: Programming and Problem Solving
Course Code: 0705210206

Batch: 2022-26 (R) and 2021-25 (B) (Sem II)
Regular and Backlog April/Even 2023
Examination

Date: 17/07/2023

Day: Monday

Maximum Marks: 30

Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. **Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

Q.1 a) **Design a flowchart and an algorithm to find greatest of two numbers.** **3** **CO1, BL3**

b) **Illustrate concept of computational thinking with real life example.** **3** **CO1, BL2**

Q.2 a) **The class has student roll numbers from 1 to 20.** **4** **CO2, BL3**

G1={1,2,3,4,5,6,7,8,9}
G2={5,6,7,8,9,10,11,12,13}
G3={1,2,6,7,11,12,15,16,20}
G1 is a set of roll numbers who play cricket.
G2 is a set of roll numbers who play football.
G3 is a set of roll numbers who play chess.
Write a python program to answer the following questions related to above mentioned sets:
(i) Display the roll number of students who play outdoor games only.
(ii) Display the roll number of students who play all the games.

b) **In a multi national company, the HR wants to keep the records of recruited employees. He has to maintain the data as id, and name, age, salary, address. If new employee is recruited he should not get the same emp id. Write a python program to add new employee and retrieve all the data of the existing employees based on his id. (Hint: Use dictionary)** **4** **CO2, BL3**